ANALYSTLAB AFRICA — DATA SCIENCE INTERNSHIP

# SwiftLog Logistics Analytics & AI Transformation

## Week 5 Technical Report — Supply Chain Intelligence & Feature Engineering for Predictive Logistics

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Data Science Track | AnalystLab Africa Experience Lab — Batch E

| Document control | Detail |
|---|---|
| Week | Week 5 |
| Track | Data Science |
| Mission | DS05 — Supply Chain Intelligence & Feature Engineering for Predictive Logistics |
| Intern | Takudzwa Jeffrey Gumbo |
| Project | SwiftLog Logistics Analytics & AI Transformation |
| Submission deadline | 18 October 2026, 11:59 PM WAT |
| Data period | 1 January 2026 — 8 September 2026 |
| Core analytical dataset | 5,000 orders; 4,764 with order-item/revenue coverage |
| Key limitation | Supplier table has no operational join key; warehouse capacity absent |

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# 1. Executive Summary

This Week 5 Data Science engagement integrates SwiftLog's operational data into a reproducible analytical foundation for future predictive logistics. The final order-level dataset preserves all 5,000 orders and links customer, shipment, delivery and warehouse information; 4,764 orders have product-level item records from which revenue and basket features can be calculated, while 236 orders are retained and explicitly flagged as lacking item detail.

| Headline measure | Result | Business interpretation |
|---|---|---|
| Revenue represented | $21,668,065.85 | Calculated from available order-item quantities × product prices |
| Average observed order value | $4,548.29 | Across orders with order-item coverage |
| Delayed-delivery rate | 63.7% | Primary operational concern; nearly two thirds labelled delayed |
| Average delay | 3.57 days | Observed across all 5,000 shipments |
| Returned-order rate | 9.5% | Order-level outcome; product-level return rate unavailable |
| Largest revenue category | Fashion — $8,059,602.65 | Fashion contributes the largest category revenue |
| Peak order month | August — 654 orders | Capacity planning should reflect this demand peak |
| Warehouse delay range | 58.7% to 68.0% | Observed spread, but H1 finds no statistically significant distributional difference |

- All three formal hypothesis tests failed to reject their null hypotheses: warehouse delay differences, delay differences for returned versus non-returned orders, and city-level revenue distribution differences were not statistically supported.
- The strongest customer segment by total revenue is High-Value Loyal; product segmentation separates revenue-and-volume leaders, premium-revenue products, volume drivers and long-tail items.
- Supplier performance cannot be calculated responsibly because SupplierID exists only in the standalone supplier table. No mapping was fabricated.
- Warehouse utilisation and inventory turnover are reported only as clearly named proxies because capacity and time-stamped inventory snapshots are missing.

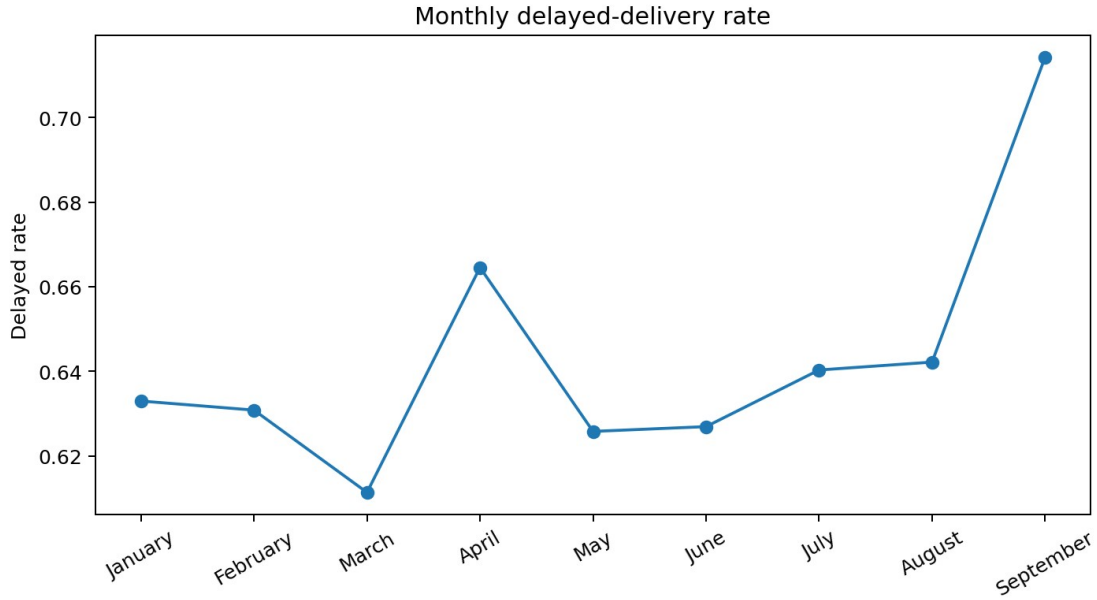



*Figure 1 — Monthly delayed-delivery rate. September is a partial month and should be interpreted cautiously.*

# 2. Business Context and Analytical Objectives

SwiftLog is a growing logistics and supply-chain business seeking greater visibility into delivery efficiency, customer value, inventory, warehouse operations and product performance. The Data Science mission is deliberately upstream of predictive modelling: the goal is to integrate imperfect operational data, identify patterns and limitations, engineer meaningful features and deliver an analytical dataset suitable for later Machine Learning work.

| Objective | How this submission addresses it |
|---|---|
| Integrate data | Validated joins across customers, orders, items, products, shipments, delivery, warehouses, employees and returns. |
| Understand operations | 21+ professional figures and descriptive statistics across demand, revenue, delivery, inventory, returns and staffing. |
| Test business hypotheses | Three non-parametric inferential tests with effect sizes and business interpretation. |
| Engineer predictive features | Order value/size, prior customer history, seasonality, warehouse load proxies, delay/return outcomes and inventory-turnover proxy. |
| Discover segments | Exploratory K-Means segmentation for customers, products and warehouses. |
| Prepare for ML | Self-contained Colab notebook plus exported order-level feature dataset and explicit leakage/availability notes. |

# 3. Data Acquisition & Understanding

## 3.1 Dataset portfolio

| Dataset | Rows | Cols | Missing cells | Exact duplicate rows |
|---|---|---|---|---|
| Customers | 1,000 | 3 | 0 | 0 |
| Orders | 5,000 | 9 | 0 | 0 |
| Order_Items | 15,000 | 4 | 0 | 0 |
| Products | 250 | 3 | 0 | 0 |
| Suppliers | 150 | 2 | 0 | 0 |
| Warehouses | 15 | 2 | 0 | 0 |
| Inventory | 4,000 | 3 | 0 | 3 |
| Shipments | 5,000 | 4 | 0 | 0 |
| Delivery_Performance | 5,000 | 3 | 0 | 0 |
| Employees | 200 | 2 | 0 | 0 |
| Returns | 500 | 3 | 0 | 0 |
| Data_Dictionary | 11 | 2 | 0 | 0 |

All supplied CSVs have zero missing cells, but structural completeness issues remain. Most importantly, 236 Orders have no matching Order_Items record. Referential integrity checks show that every child foreign key that is present resolves to a valid parent; this does not imply every parent record has children.

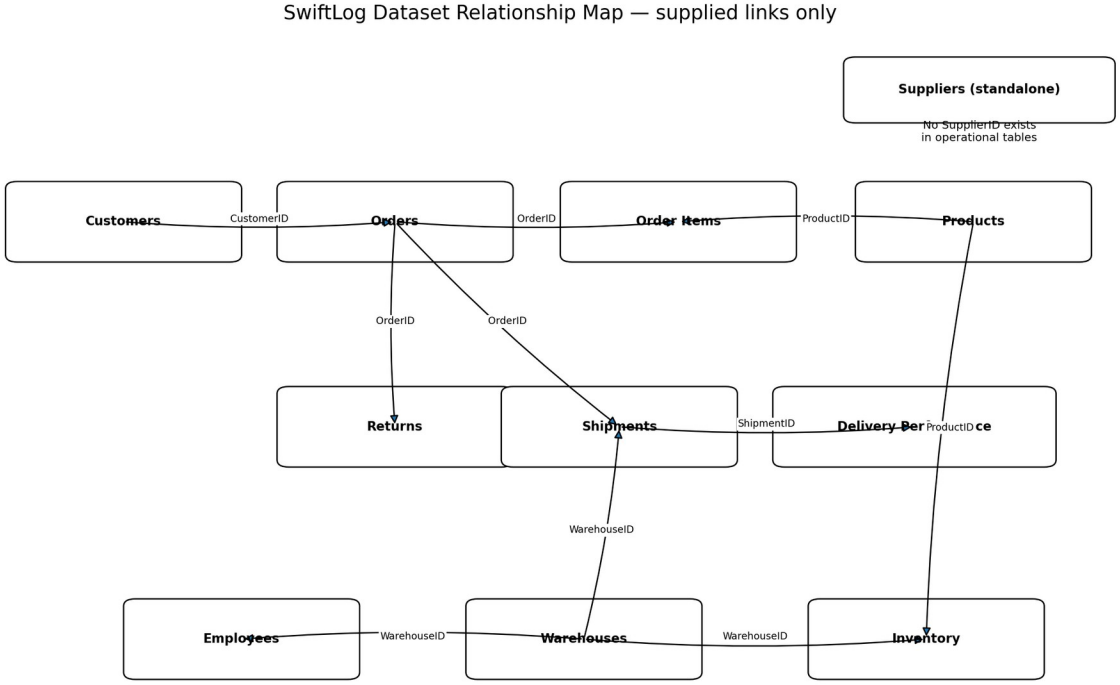



*Figure 2 — Validated SwiftLog relationship map. Suppliers remain standalone because no SupplierID exists elsewhere.*

## 3.2 Critical structural findings

| Finding | Evidence | Treatment |
|---|---|---|
| Supplier disconnect | SupplierID occurs only in SwiftLog_Suppliers.csv | Keep table standalone; supplier reliability not calculated. |
| Order-item coverage gap | 236 of 5,000 Orders have no Order_Items | Retain orders; mark OrderItemDataAvailable=0; exclude missing revenue only from revenue-specific statistics. |
| Inventory repetition | 1,545 repeated Warehouse–Product rows beyond first; 3 exact repeated rows | Retain as stock movements per official guidance; aggregate separately for reporting. |
| No warehouse capacity | Warehouses contains WarehouseID + Location only | Use explicitly labelled load proxies, never true utilisation. |
| Order-level returns | Returns contains ReturnID, OrderID, Reason only | Do not claim true product return rates; use returned-order exposure proxy. |
| Simplified dictionary | Dataset names only | Infer cautiously and record every assumption. |

# 4. Data Cleaning & Preprocessing

Cleaning was intentionally conservative. No row was deleted simply because it looked repetitive. Strings were trimmed, OrderDate was parsed to datetime, calendar features were derived, positive/non-negative range assertions were applied, Shipment and Delivery_Performance delay values were cross-checked, and inventory movement rows were preserved.

| Step | Action | Result |
|---|---|---|
| Text standardisation | Trim whitespace from object fields | Consistent categorical matching without recoding business values. |
| Date handling | Parse OrderDate; derive year, month, | Supports demand/seasonality analysis. |

| | | |
|---|---|---|
| | month name, quarter, day of week | |
| Range validation | Quantity/price > 0; stock/delay >= 0 | All checks passed. |
| Delay cross-check | Compare Shipments.DelayDays with Delivery_Performance.DelayDays by ShipmentID | No mismatches. |
| Inventory treatment | Preserve all movement records; aggregate Warehouse–Product view separately | 2,455 aggregated Warehouse–Product combinations from 4,000 movement rows. |
| Return treatment | Aggregate multiple ReturnID records to order-level HasReturn / ReturnCount | 473 unique returned orders across 500 return records. |

# 5. Exploratory Data Analysis

More than the required 15 visualisations were generated. The report highlights the most decision-relevant charts; the full figure set is stored under Visualizations/Charts and reproduced in the notebook.

## 5.1 Demand and revenue

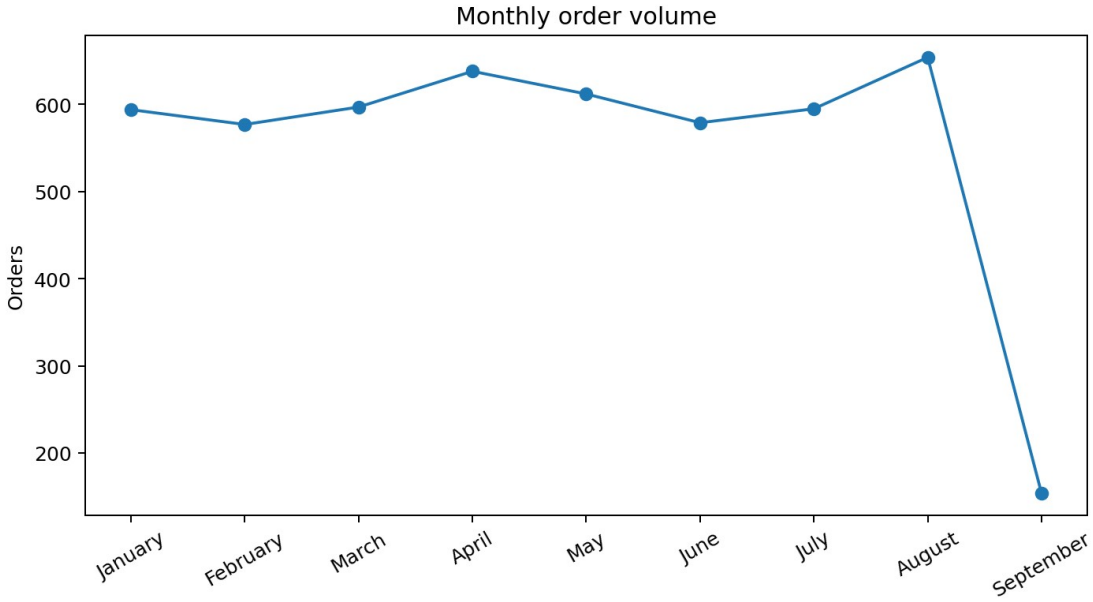



*Figure 3 — Monthly order volume. August is the peak month with 654 orders; September is partial.*

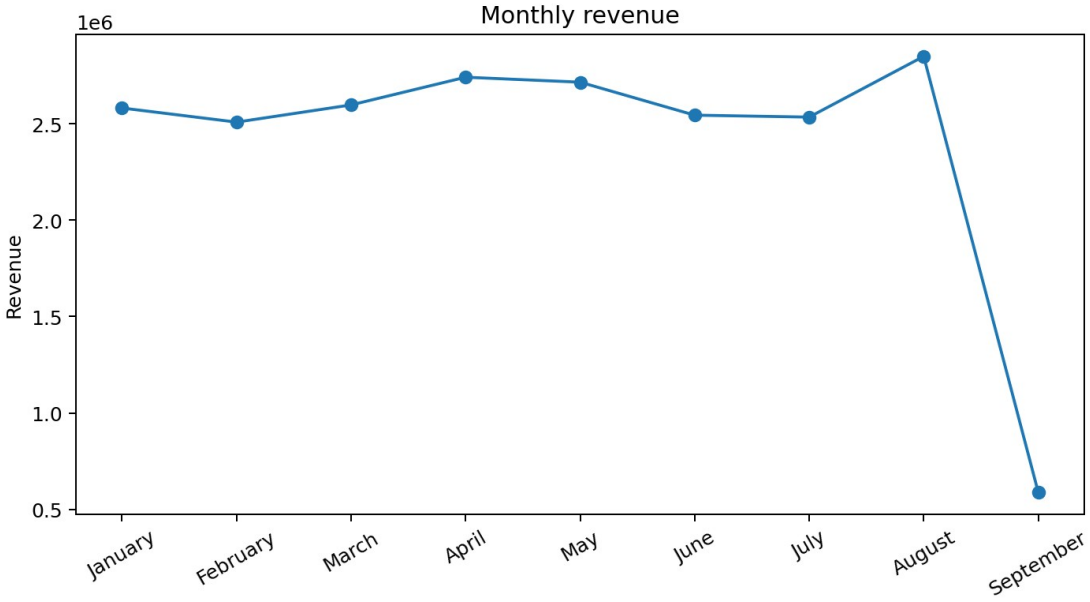



*Figure 4 — Monthly observed revenue. August is also the highest-revenue month.*

Observed order-item revenue totals $21,668,065.85. The top category is Fashion at $8,059,602.65; Food and Electronics follow. Because 236 orders lack items, this is observed revenue from the available line-item data rather than a guaranteed complete ledger total.

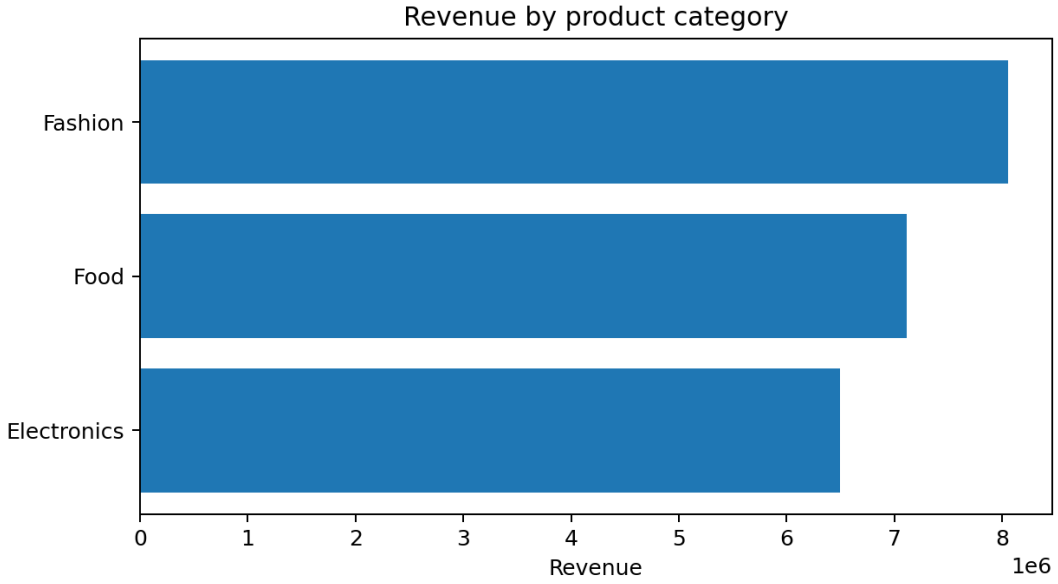



*Figure 5 — Revenue by product category.*

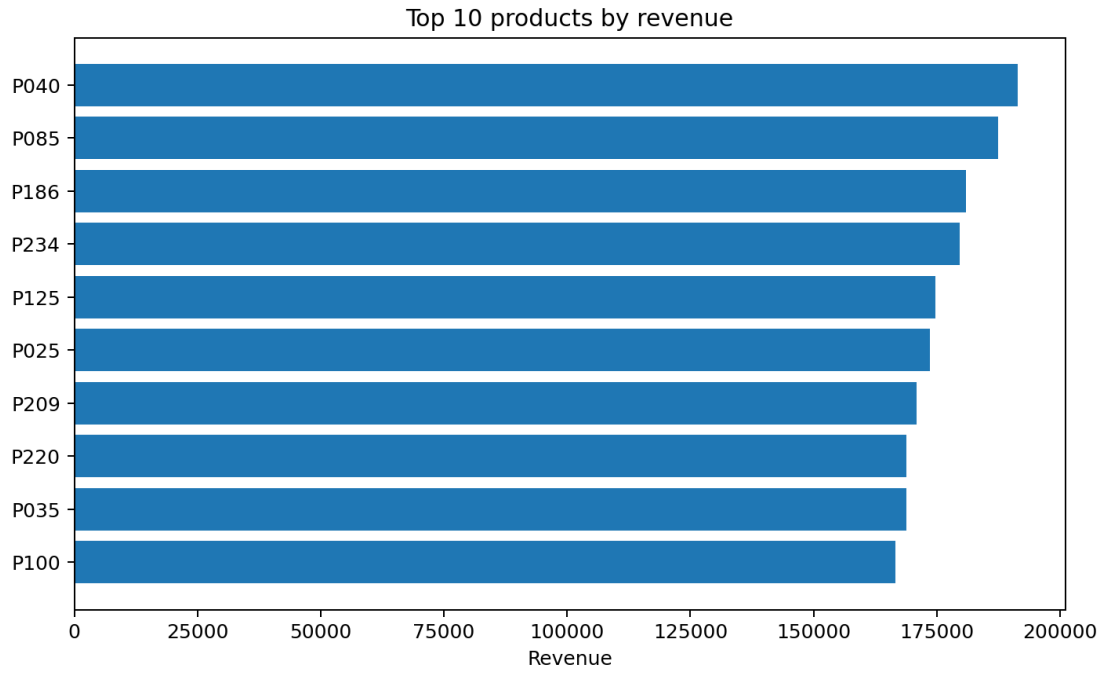



*Figure 6 — Top 10 products by revenue. Product P040 leads the observed product portfolio.*

## 5.2 Delivery and warehouse operations

The overall delayed-delivery rate is 63.7%, with average delay of 3.57 days. W05 has the highest observed delayed rate (68.0%) while W01 has the lowest (58.7%); however, inferential testing later shows the warehouse delay distributions are not significantly different.

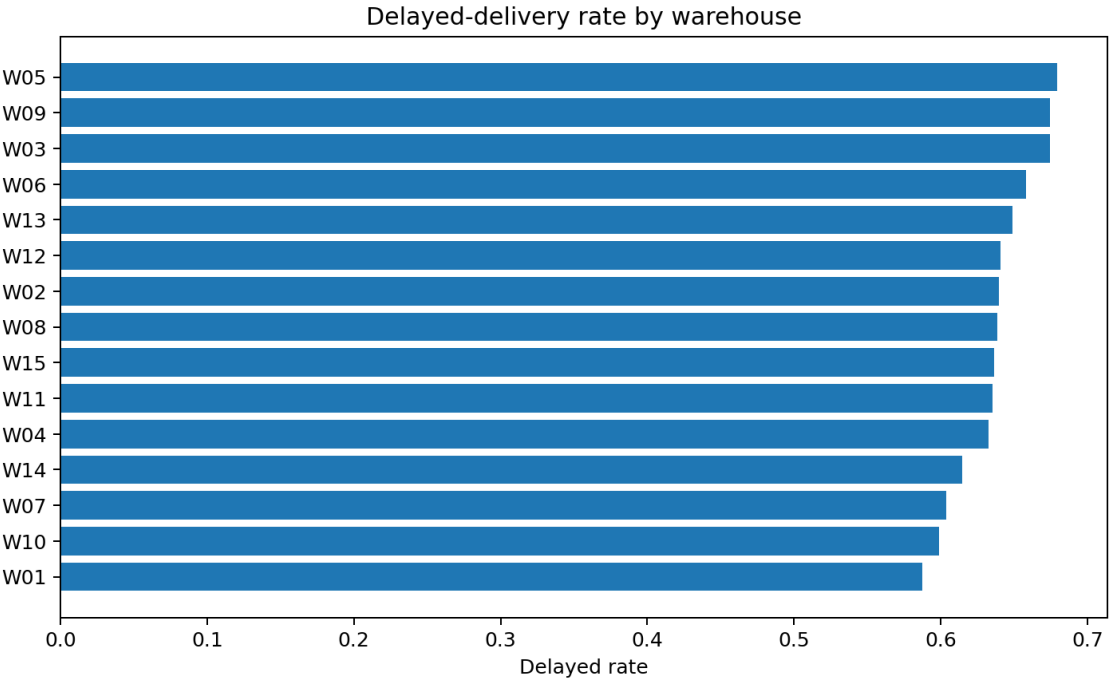



*Figure 7 — Observed delayed-delivery rate by warehouse.*

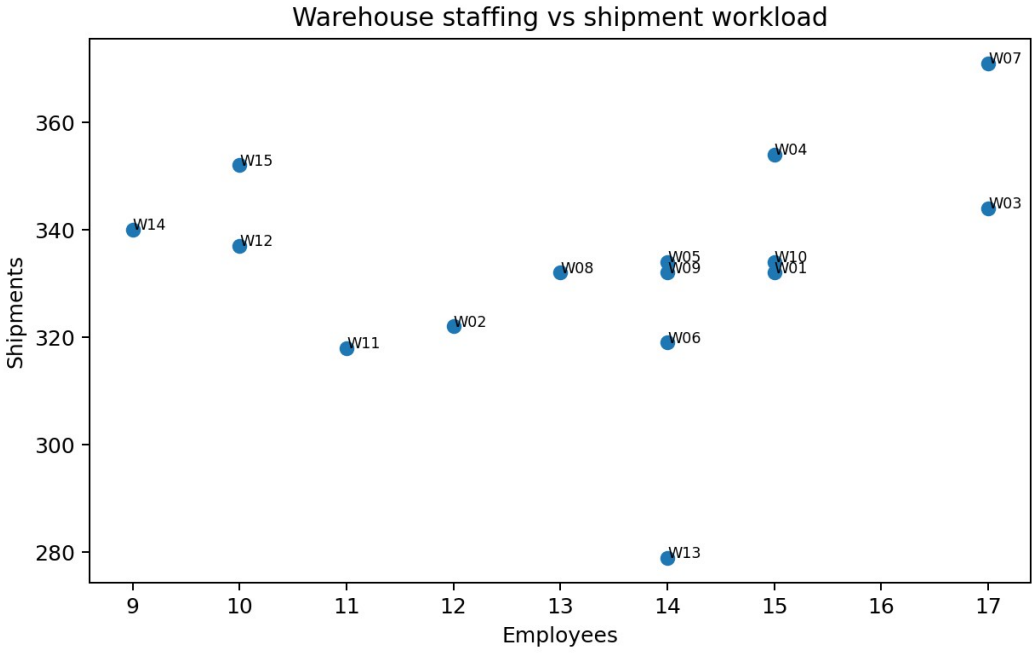



*Figure 8 — Staffing vs shipment workload. This is descriptive because staffing productivity standards are not supplied.*

## 5.3 Returns and inventory

The order-level return rate is 9.5%. Late Delivery is the most common stated return reason (185 records), followed by Wrong Item (163) and Damaged (152). Returns cannot be assigned to a specific product because ProductID/OrderItemID is absent from Returns.

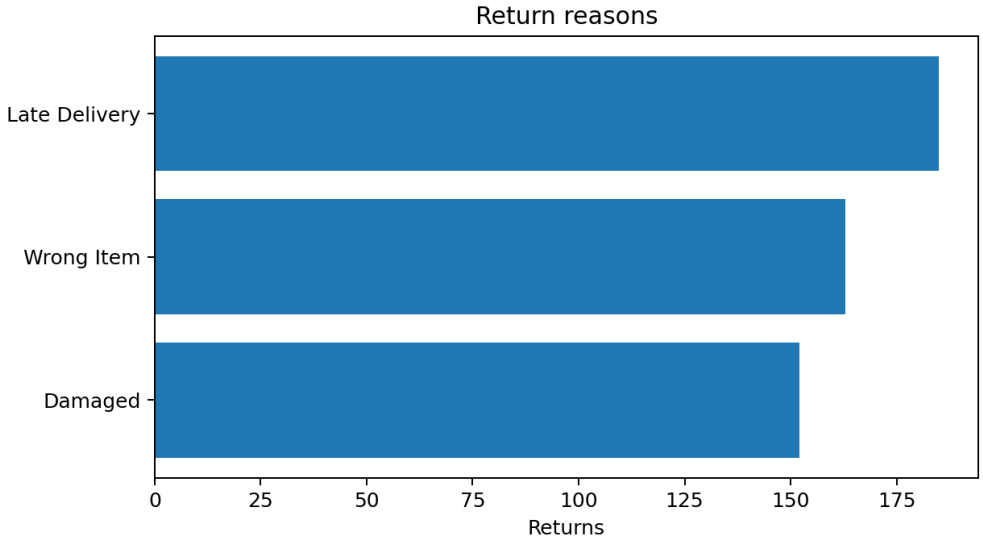



*Figure 9 — Return reasons.*

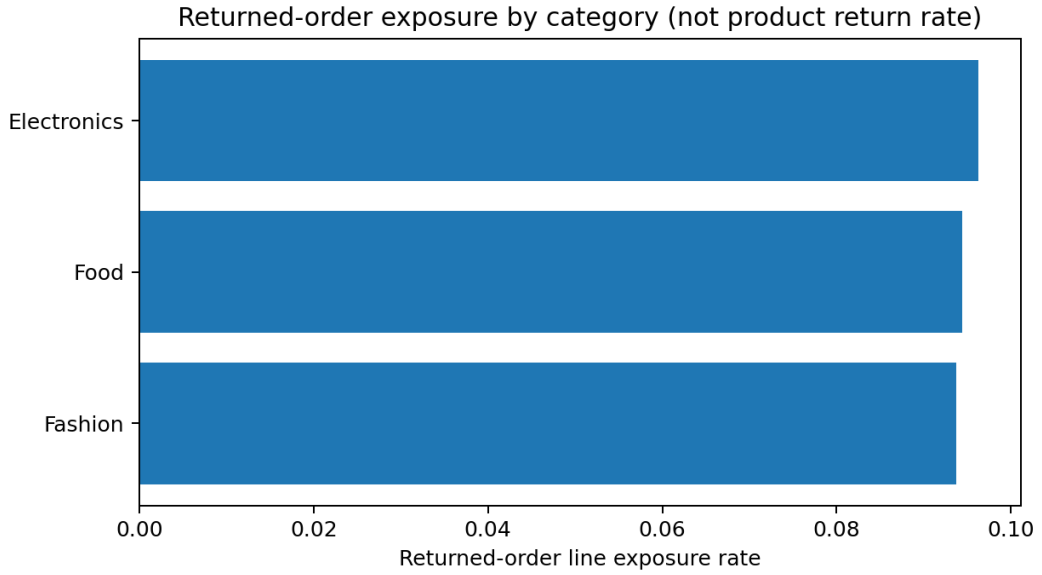



*Figure 10 — Category exposure to returned orders. This is not a true product return rate.*

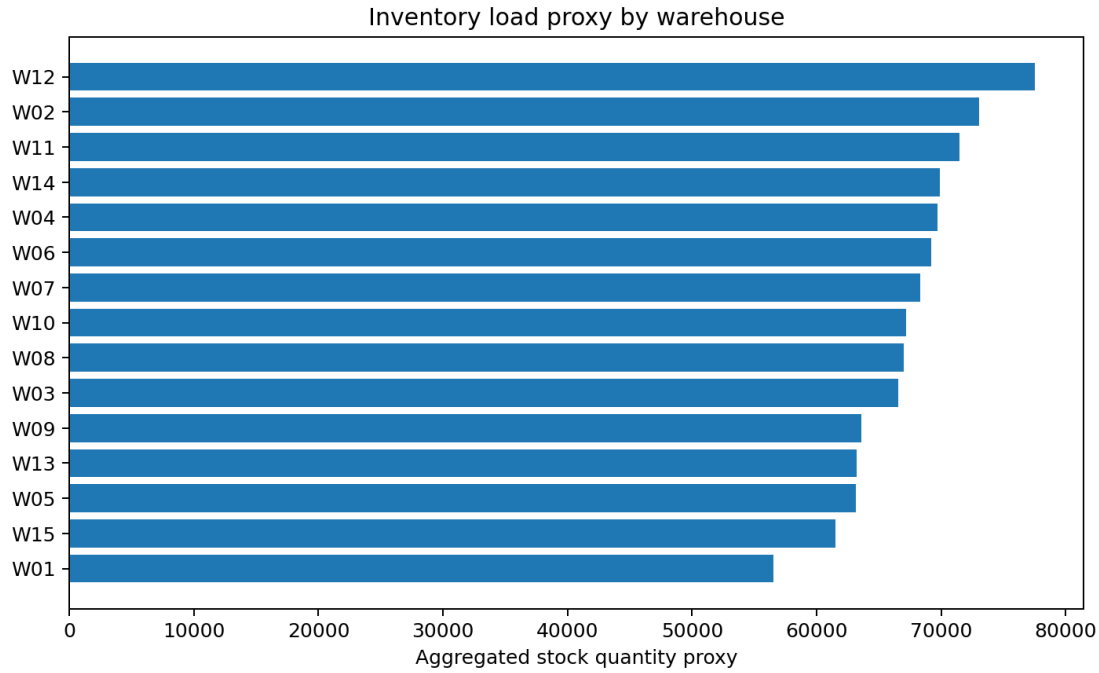



*Figure 11 — Warehouse inventory-load proxy using summed stock movements. True capacity utilisation cannot be computed.*

# 6. Statistical Analysis

Non-parametric tests were selected because delay days are discrete and order revenue is highly skewed. The project keeps unsupported hypotheses visible as results rather than quietly discarding them.

| ID | Test | Statistic | p-value | Effect size | Decision |
|---|---|---|---|---|---|
| H1 | Kruskal-Wallis | 10.039 | 0.7594 | 0.0000 | Fail to reject |
| H2 | Mann-Whitney U | 1069995.000 | 0.9828 | 0.0006 | Fail to reject |
| H3 | Kruskal-Wallis | 3.432 | 0.3297 | 0.0001 | Fail to reject |

- H1 — Delivery delays across warehouses: fail to reject (p=0.7594). The visible rate spread is not statistically strong enough to conclude warehouse-specific delay distributions differ.

- H2 — Returned vs non-returned orders: fail to reject (p=0.9828). Overall delivery delay does not distinguish returned orders from non-returned orders in this synthetic sample.
- H3 — Order revenue across customer cities: fail to reject (p=0.3297). City revenue totals are not evidence of different per-order revenue distributions.

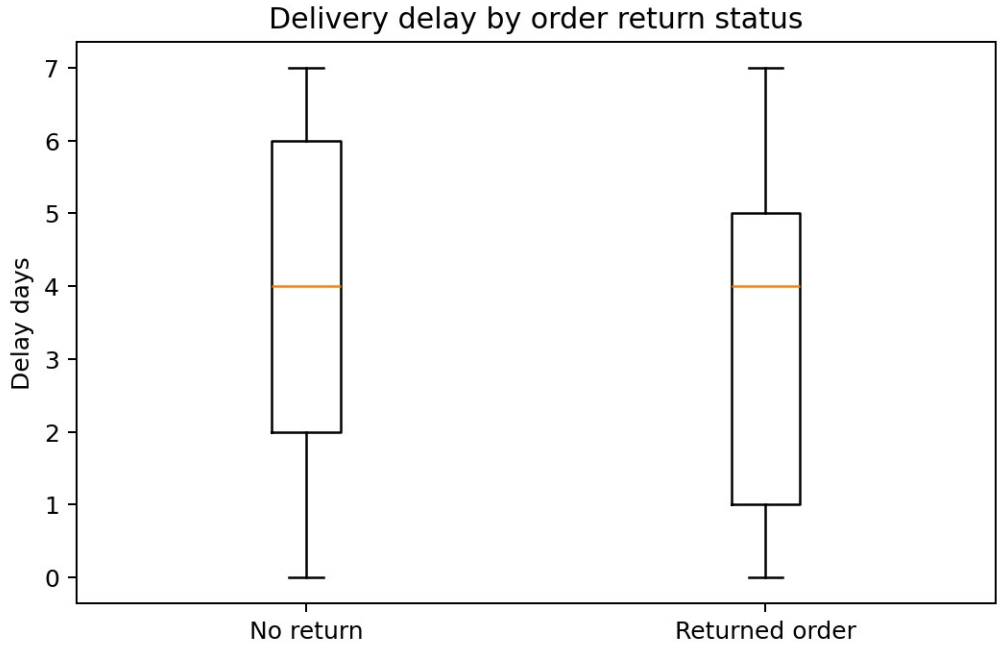



*Figure 12 — Delivery delay by order return status; the distributions overlap strongly.*

# 7. Feature Engineering & ML Readiness

The final feature dataset is order-level and preserves all 5,000 orders. Outcome-derived variables such as IsDelayed, DelayDays and DeliveryEfficiencyScore are included for analysis/target definition but must not be used as predictors when modelling delay. Historical customer features are constructed from prior orders only to reduce temporal leakage risk.

| Feature / feature group | Construction | Business / modelling value | Caution |
|---|---|---|---|
| OrderRevenue | Σ Quantity × UnitPrice by OrderID | Order value and customer value modelling | Missing for 236 orders without items. |
| TotalUnits / DistinctProducts / UniqueCategories | Order-level item aggregation | Basket size and complexity | Unavailable where Order_Items absent. |
| PriorCustomerOrders / PriorCustomerSpend | Strict cumulative history before each order | Customer familiarity/value without future orders | Revenue history depends on available item coverage. |
| SeasonalDemandIndicator | Monthly order volume above median | Peak-demand planning | Descriptive; full-period threshold should be re-fit inside ML training. |
| WarehouseOperationalLoadProxy | 50% normalized stock proxy + 50% normalized shipment volume | Operational workload context | Not true utilisation; no capacity. |
| InventoryTurnoverProxy | Units shipped / aggregated stock movement quantity | Relative stock-demand signal | Not accounting turnover; no dated inventory/average stock. |
| DeliveryEfficiencyScore | 100 × (1 - DelayDays/max delay) | Descriptive operational index | Outcome-derived; never use to predict delay. |
| HasReturn / ReturnCount | Order-level aggregation of Returns | Potential return-modelling target | Cannot identify returned product. |

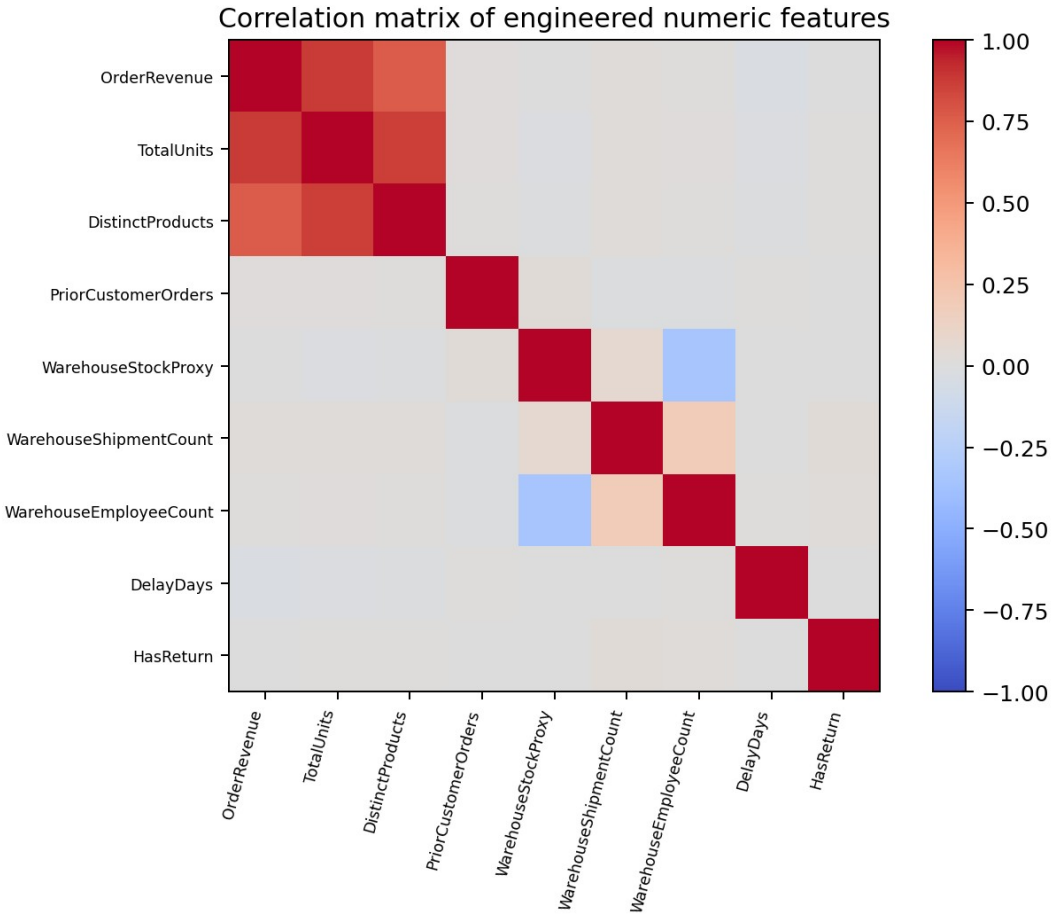



*Figure 13 — Correlation matrix of selected engineered numeric features.*

# 8. Segmentation & Pattern Discovery

## 8.1 Customer segmentation

| Segment | Customers | Total revenue | Avg revenue | Avg orders | Avg order value |
|---|---|---|---|---|---|
| High-Value Loyal | 329 | $11,480,070 | $34,894 | 7.58 | $4,846 |
| High-Value Occasional | 246 | $5,398,755 | $21,946 | 3.77 | $6,210 |
| Core / Lower-Value | 416 | $4,789,241 | $11,513 | 3.80 | $3,183 |

Customer clustering uses value/frequency variables only and achieves silhouette 0.360. This is moderate exploratory separation, not evidence of naturally discrete customer classes. High-Value Loyal customers contribute the largest segment revenue.

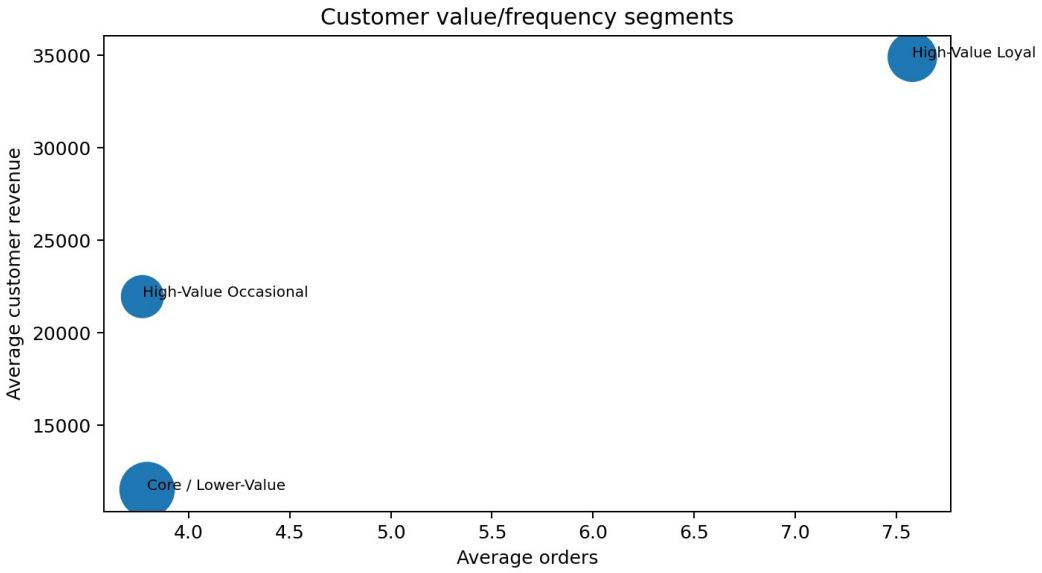



*Figure 14 — Customer value/frequency segments.*

## 8.2 Product and warehouse patterns

Product segmentation silhouette is 0.330; warehouse segmentation silhouette is 0.301. Both are exploratory, particularly warehouse clustering where only 15 warehouses exist.

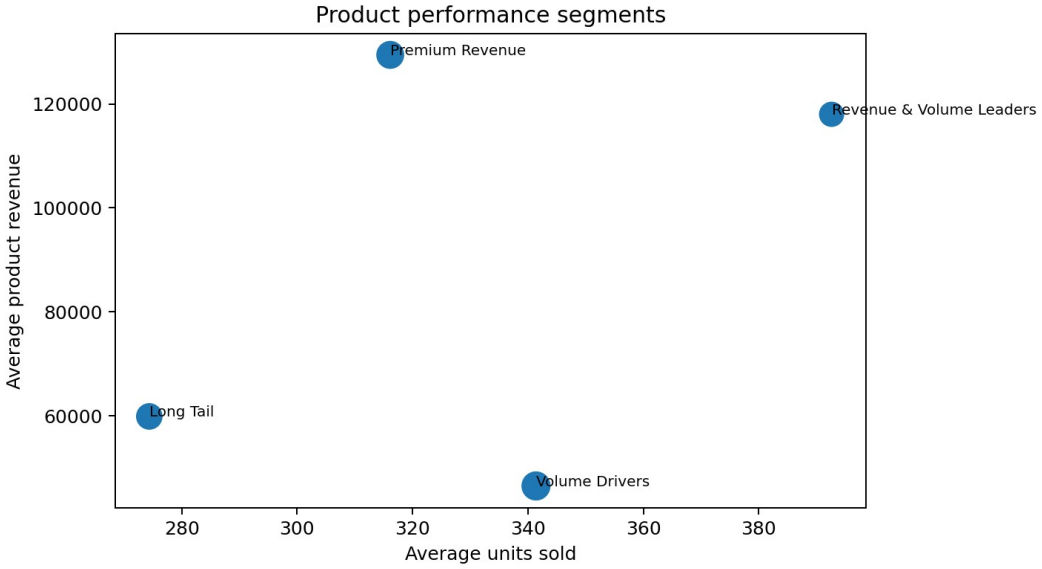



*Figure 15 — Product performance segments.*

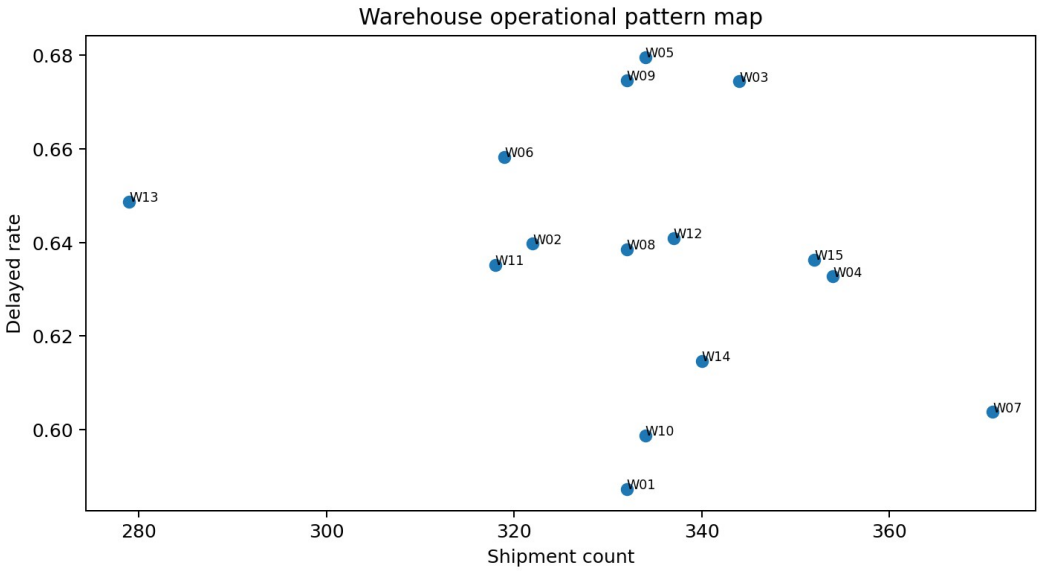



*Figure 16 — Warehouse operational pattern map.*

# 9. Business Insights — Assignment Questions

**1. Which customer segments generate the highest business value?**

High-Value Loyal customers lead total revenue at approximately $11.48m across 329 customers, with the highest average order frequency.

**2. Which products contribute most to revenue?**

P040 is the highest-revenue individual product at $191,463.68; Fashion is the highest-revenue category at $8,059,602.65.

**3. Which suppliers consistently underperform?**

Not determinable from the provided data. Suppliers has no operational join key; creating supplier scores would require an invented mapping, which this submission does not do.

**4. Which warehouses require operational improvements?**

W05, W09 and W03 have the highest observed delayed rates, with W05 at 68.0%. However H1 p=0.759 means this ranking is a monitoring priority, not statistical proof of warehouse-caused delay.

**5. Which products have the highest return rates?**

Not determinable exactly because Returns is order-level. Electronics has the highest returned-order line exposure proxy (~9.6%), but this must not be called a true product return rate.

**6. Which regions experience the greatest delivery delays?**

Using warehouse location as the geographic proxy, Lagos has the highest average delay (~3.59 days), slightly above Kano and Abuja.

**7. What operational bottlenecks were identified?**

The strongest operational signal is the 63.7% delayed-delivery rate. Data-system bottlenecks include the supplier disconnect, missing warehouse capacity, undated inventory movements, order-level-only returns and 236 orders without item records.

**8. Which variables are most suitable for predictive modelling?**

Order size/value, product mix, calendar/seasonal features, prior customer history, warehouse/location, staffing and load proxies. Delay-derived fields are targets/outcomes and must be excluded from delay predictors.

### 9. What hidden patterns did the analysis uncover?

Warehouse delay differences and return-related delay differences look plausible descriptively but are statistically unsupported. August is the demand/revenue peak; Fashion leads revenue; customer value segmentation is more distinct than warehouse segmentation.

### 10. What recommendations improve operations?

Prioritise delay reduction, improve data capture, protect high-value customers, use product segments for inventory prioritisation, prepare for August demand, and collect supplier/capacity/return-line data before advanced optimisation.

# 10. Strategic Recommendations

| Area | Recommendation |
|---|---|
| Delivery optimisation | Create a focused delay-reduction programme and track delayed rate/average delay weekly; the overall delayed rate is 63.7%. |
| Warehouse process review | Investigate W05/W09/W03 first while treating observed rankings as diagnostic priorities rather than causal proof. |
| Benchmarking | Review processes at W01/W10/W07 as lower-delay operational references. |
| Peak planning | Increase capacity planning and staffing review ahead of August, the highest order-volume and revenue month. |
| Customer retention | Protect High-Value Loyal customers with service-level monitoring and targeted retention because they contribute the largest segment revenue. |
| Category strategy | Protect Fashion availability while balancing concentration risk across Food and Electronics. |
| Product portfolio | Use product segments to define differentiated replenishment and merchandising rules for leaders, premium products, volume drivers and long-tail items. |
| Returns | Treat Late Delivery as a priority complaint reason, but improve return data before attributing product-level quality problems. |
| Supplier data governance | Add SupplierID to product/contract/shipment records so supplier reliability can be measured rather than assumed. |
| Warehouse data governance | Capture true capacity, space used and dated inventory snapshots so true utilisation and turnover can replace proxies. |
| Returns granularity | Add ProductID/OrderItemID and return quantity to Returns to enable valid product/category return rates. |
| Order completeness | Resolve the 236 orders missing Order_Items and add automated referential/completeness controls in the source system. |

# 11. Assumptions, Data Limitations & Risks

| Area | Limitation / assumption | Impact | Mitigation |
|---|---|---|---|
| Suppliers | No SupplierID outside Suppliers | Supplier performance analysis not possible | Keep standalone; recommend source-system mapping. |
| Warehouse utilisation | No capacity metric | True utilisation cannot be computed | Use labelled load proxies only. |
| Inventory | No date/current-record indicator | Stock movements cannot form time-series inventory | Aggregate for reporting; call turnover a proxy. |
| Returns | No ProductID/OrderItemID | Cannot determine returned product | Use order-level return analysis and exposure proxy. |
| Order items | 236 Orders have no Order_Items | Observed revenue is incomplete for those orders | Retain orders and flag coverage; avoid zero imputation. |
| Data dictionary | No field-level definitions | Business meanings partly inferred | Document rules; recommend governed dictionary. |
| September | Data ends 8 September | September monthly figures are partial | Do not compare September totals directly with full months. |

| | | | |
|---|---|---|---|
| Segmentation | Unsupervised clusters depend on chosen features/k | Segments are exploratory | Report silhouette and avoid causal claims. |

# 12. Testing, QA & Decision Log

| Work completed | Test / evaluation | Finding | Action | Result |
|---|---|---|---|---|
| Dataset integration | Foreign-key anti-joins | Zero unmatched child keys | Retain validated relationships | Join integrity confirmed. |
| Order completeness | Parent-to-child coverage check | 236 Orders lack Order_Items | Use left merge + coverage flag | All 5,000 orders preserved. |
| Shipment/delivery consistency | Compare DelayDays by ShipmentID | No mismatches | Use delivery table status and validated delay | One consistent delay measure. |
| Inventory cleaning | Check duplicate and Warehouse–Product repetition | Repeated rows are expected movements per guidance | Retain; aggregate separately | No incorrect de-duplication. |
| Hypotheses | Kruskal-Wallis / Mann-Whitney | All three nulls fail to reject | Report as findings | No p-value fishing. |
| Segmentation | Silhouette scoring | Customer separation strongest; warehouse/product moderate | Treat clusters as exploratory | No overclaiming. |
| Notebook reproducibility | Clean end-to-end execution | All cells completed successfully | Keep self-contained Drive file search | Colab-ready submission. |

## 12.1 Decision log

| Decision | Reason | Evidence | Impact |
|---|---|---|---|
| D1: do not fabricate supplier mapping | No SupplierID relationship | Official AnalystLab guidance + schema | Supplier scoring omitted honestly. |
| D2: retain inventory repeats | They represent movements | Official AnalystLab guidance | No false data loss. |
| D3: preserve itemless orders | Delivery/customer evidence remains valid | 236 parent orders lack child lines | 5,000-order analytical grain preserved. |
| D4: label warehouse metrics as proxies | Capacity absent | Warehouse schema contains only ID and location | Prevents false utilisation claims. |
| D5: use non-parametric tests | Skewed/discrete variables | Revenue and DelayDays distributions | Robust statistical inference. |
| D6: keep unsupported hypotheses | Null findings are informative | p-values 0.759, 0.983, 0.330 | Avoid cherry-picking. |

# 13. Future Machine Learning Opportunities

| Opportunity | Potential target | Candidate predictors | Precondition |
|---|---|---|---|
| Delivery-delay prediction | IsDelayed / DelayDays | Order size, product mix, month, prior customer history, warehouse/location/load | Do not use DelayDays, DeliveryStatus or DeliveryEfficiencyScore as predictors. |
| Demand forecasting | Monthly/category/product units | Historical quantities, seasonality, product/category | Longer time series and dated inventory recommended. |
| Return-risk modelling | HasReturn | Order/product mix, customer history, delivery context | Add returned ProductID/OrderItemID for product-level modelling. |
| Supplier reliability | Future supplier outcome | Supplier, product, shipment, delay | Requires SupplierID mapping. |
| Warehouse optimisation | Delay/load/service metrics | Capacity, staffing, inventory, shipments | Requires true capacity and dated stock snapshots. |

# 14. Conclusion

The SwiftLog analysis delivers a defensible, reproducible foundation for later Machine Learning without pretending that imperfect data is complete. The strongest operational issue is delivery performance, while the strongest data-governance gaps are supplier disconnection, absent warehouse capacity, undated

inventory and order-level-only returns. Revenue and demand patterns identify clear priorities — August peak planning, protection of high-value customers and product/category portfolio management — while statistical tests prevent descriptive differences from being overstated. The final feature-engineered dataset, notebook, figures, profiling report and documentation provide a transparent handoff for future predictive logistics work.

# 15. Appendices

## 15.1 File manifest

- Data/Original_Datasets/ — 12 supplied CSVs
- Data/Cleaned_Datasets/ — cleaned tables + aggregated inventory
- Data/Feature_Engineered_Dataset.csv — 5,000-row order-level analytical dataset
- Notebook/Week5_DataScience.ipynb — self-contained executable notebook
- Reports/Data_Profiling_Report.html
- Reports/Technical_Report.pdf / .docx
- Reports/Executive_Summary.pdf / .docx
- Visualizations/Charts/ — 20+ generated figures
- Presentation/Executive_Presentation.pptx
- README.md, WEEK5_SUBMISSION_CHECKLIST.md, social_posts_week5.md, requirements.txt

## 15.2 Reproducibility statement

All numerical claims in this report trace to code executed against the supplied CSV files. The notebook searches recursively under /content/drive/MyDrive/Analyst Lab Africa when run in Colab, reproduces the cleaning, integration, analysis, hypothesis tests, feature engineering and segmentation, and exports the resulting artifacts. Random seed 42 is fixed for K-Means segmentation.

## 15.3 Responsible interpretation

This is an educational consulting project using supplied synthetic/fictional business data. Proxies are labelled as proxies; unsupported relationships are not fabricated; statistical associations are not presented as causal conclusions; and future Machine Learning opportunities are recommendations rather than claims of deployed predictive performance.